# Coding & Solution

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| **Date** | 06 July 2026 |
| **Team ID** | |
| **Project Name** | Ledgerline – Credit Card Approval Prediction System |
| **Maximum Marks** | 5 Marks |

## Solution Summary

| Field | Details |
|---|---|
| Repository Link / URL | <Paste your GitHub repository link here> |
| Programming Language(s) | Python |
| Framework(s) Used | Flask, scikit-learn, XGBoost, pandas, NumPy, matplotlib, Jupyter |
| Key Features Implemented | Single-applicant prediction form • JSON API endpoint • Batch CSV screening with downloadable results • 4-model comparison & calibrated decision threshold • Model transparency report (ROC curves, feature importance) |
| Pending / Incomplete Features | IBM Watson Machine Learning live deployment (script is written and ready, but not yet run against live IBM Cloud credentials) • User authentication / role-based access |
| Setup / Run Instructions | 1. pip install -r requirements.txt 2. python app.py 3. Open http://127.0.0.1:5000 in a browser (Full instructions in README.md) |

## Code Quality Checklist

| S.No | Criteria | Status (Yes/No) |
|---|---|---|
| 1 | Code is modular and organized into functions / classes | Yes |
| 2 | Meaningful variable and function names are used | Yes |
| 3 | Code includes comments / documentation where necessary | Yes |
| 4 | Error handling is implemented for critical operations | Yes |
| 5 | The application runs without critical errors | Yes |
| 6 | Code is committed to a version control repository | Yes |

## Additional Notes / Comments

The app's preprocessing logic (build_feature_row) keeps the exact one-hot column order used during training in a single source of truth (feature_columns.json), so the Flask app and the training pipeline can never drift out of sync.